Underwater Vessel Noise Classification Using the ShipsEar Dataset: Background, Acoustic Theory, and Formal Problem Definition

Somya Pratap Singh
500124413
Akshat Choudhary
500124119
Kapil Rao
500119132

***Abstract*—This report establishes the scientific and engineering context for underwater vessel noise classification using the ShipsEar dataset [1]. We review the dataset's recording protocol and hydrophone specifications, examine the physical mechanisms governing underwater acoustic propagation (sound speed, transmission loss, multipath interference, propeller cavitation, machinery harmonics, and blade-rate modulation), characterise the five vessel categories in terms of their distinct spectral signatures, and formally define the classification task as a supervised multi-class pattern recognition problem. This background forms the theoretical foundation for all subsequent feature extraction, model training, and evaluation stages of the project.**

***Index Terms*—underwater acoustics, vessel noise classification, ShipsEar, propeller cavitation, LOFAR, DEMON, hydrophone, multi-class classification**

I. Introduction

Identifying a vessel from its acoustic signature — without visual contact — is a long-standing problem in naval sonar, maritime surveillance, and marine environmental monitoring. A submerged hydrophone records a composite signal arising from propeller cavitation, engine vibration, hull flow noise, and propeller blade-rate modulation. Each vessel type imprints a characteristic spectral fingerprint on this signal, making automatic classification feasible through signal processing and machine learning.

The ShipsEar dataset [1], introduced by Santos-Dom´ınguez *et al.* in 2016, provides the first publicly available benchmark corpus for this task, covering five operationally distinct vessel classes recorded in a realistic estuarine environment. This part of the project reviews the dataset, the relevant acoustic physics, and states the classification objective precisely in mathematical terms.

II. Dataset Overview: ShipsEar

A. Recording Setup

ShipsEar was recorded in the R´ıa de Vigo estuary, northwestern Spain. An omnidirectional HTI-96-MIN hydrophone was deployed at approximately 1.5 m depth. The sensor exhibits a flat frequency response from 2 Hz to 30 kHz with a sensitivity of -64 dB re 1 V/μPa, making it well-suited for capturing the full spectral bandwidth of marine vessel noise. All 90 recordings are stored as uncompressed 16-bit PCM WAV files sampled at $f_s = 52{,}000$ Hz. Recording durations vary from approximately 15 s to over 3 min, reflecting real-world operational diversity.

B. Vessel Taxonomy

The dataset is partitioned into five mutually exclusive vessel classes, summarised in Table I. The dataset is intentionally class-imbalanced: Class A (motorboats) is overrepresented relative to Classes C and E, an important consideration for classifier design and evaluation.

TABLE I
ShipsEar Vessel Classes and Acoustic Characteristics

Label	Vessel Type	Acoustic Characteristics
A	Motorboats	High-frequency broadband noise; intense propeller cavitation at high RPM; non-stationary.
B	Mussel Boats	Prominent diesel LOFAR tonal lines; slow blade-rate harmonics; stable narrowband peaks.
C	Fishing Vessels	Mixed broadband and tonal content; winch and gear noise; variable engine load.
D	Passengers / Ferries	Strong, stable engine harmonics (50–500 Hz); consistent blade-rate due to fixed routes.
E	Ocean Liners / Tugboats	Very low fundamental frequency; highest source level; complex multi-shaft harmonics.

III. Underwater Acoustic Propagation

A. Speed of Sound

Sound travels in seawater at approximately 1,500 m/s — roughly 4.4 times faster than in air (343 m/s). The empirical Mackenzie equation [4] gives the sound speed c as a function of temperature T (°C), salinity S (ppt), and depth z (m):

$$c \approx 1448.96 + 4.591T - 0.05304T^2 + 0.0002374T^3 + 1.340(S - 35) + 0.0163z \quad \text{[m/s]}. \tag{1}$$

The higher propagation speed relative to air results in longer acoustic wavelengths at any given frequency, enabling low-frequency vessel harmonics to propagate over kilometre-scale ranges with relatively low attenuation.

B. Transmission Loss

In shallow coastal waters such as the R´ıa de Vigo, acoustic energy undergoes cylindrical spreading supplemented by frequency-dependent absorption [2]:

$$\mathrm{TL} = 10\log_{10}(r) + \alpha r \quad \text{[dB]}, \tag{2}$$

where r is the source-to-receiver range in metres and α is the absorption coefficient in dB/m. Because α increases with frequency, high-frequency components (above $\sim$10 kHz) are strongly attenuated at modest ranges, concentrating discriminative vessel information below 10 kHz. This motivates resampling the ShipsEar data from 52 kHz to a lower target rate (e.g., 22,050 Hz) without information loss.

C. Multipath Interference

In shallow estuarine environments, acoustic energy undergoes multiple reflections from the pressure-release sea surface and the impedance-mismatch seabed. These multipath arrivals produce constructive and destructive interference at the hydrophone, causing frequency-selective fading and temporal smearing of tonal spectral features. Multipath effects are a principal source of within-class recording variability in ShipsEar and motivate the use of segment-level averaging in feature extraction.

IV. Vessel Noise Generation Mechanisms

Three physical mechanisms dominate underwater vessel radiated noise and directly motivate the feature extraction choices described in Part 4.

A. Propeller Cavitation

Cavitation occurs when the local pressure at a propeller blade surface falls below the vapour pressure of seawater, causing dissolved gas pockets to form and subsequently collapse. The implosions produce broadband high-frequency noise dominant above approximately 1 kHz. Cavitation intensity scales with propeller tip speed and is particularly pronounced in small, fast vessels (Class A motorboats). Mel-frequency cepstral coefficients (MFCCs) and log-mel spectrograms are effective representations for capturing this broadband energy distribution.

B. Machinery Noise and LOFAR Lines

Diesel and petrol engines generate narrowband tonal lines at integer multiples of the shaft rotation frequency. For a shaft spinning at ω revolutions per minute, the fundamental machinery frequency is:

$$f_0 = \frac{\omega}{60} \quad \text{[Hz]}. \tag{3}$$

The resulting harmonic series $\{f_0, 2f_0, 3f_0, \ldots\}$ appears as sharp peaks in a high-resolution power spectral density (PSD) estimate, commonly computed via Welch's method. These peaks are referred to as *LOFAR lines* (Low-Frequency Analysis and Recording) and are among the most class-discriminative features in ShipsEar, particularly for Classes B and D.

C. Blade-Rate Harmonics and DEMON Spectrum

As each propeller blade passes through the non-uniform wake behind a shaft strut, it generates a periodic pressure pulse. For a propeller with N_b blades on a shaft rotating at ω RPM, the fundamental blade-rate frequency is:

$$f_{\mathrm{BR}} = N_b \cdot \frac{\omega}{60} \quad \text{[Hz]}. \tag{4}$$

Blade-rate harmonics modulate the amplitude envelope of the broadband cavitation noise. The *DEMON* (Detection of Envelope Modulation on Noise) technique demodulates this envelope via a Hilbert transform, then computes the PSD of the envelope signal to reveal f_{BR} and its harmonics. DEMON features are highly discriminative for propeller-driven vessels and complement LOFAR analysis.

V. Acoustic Profiles of Vessel Classes

Building intuition for class-specific spectral characteristics is essential before modelling decisions are made.

Class A — Motorboats: Smallest and fastest vessels. High rotational speed produces intense cavitation; dominant spectral energy is broadband above 2 kHz. Signals are non-stationary due to variable throttle.

Class B — Mussel Boats: Medium-sized working vessels at slow, steady speed. Diesel LOFAR lines are sharp and stable. Blade-rate frequency is low ($\approx$1–4 Hz). The combination of strong narrowband peaks and low broadband floor makes this class relatively easy to separate.

Class C — Fishing Vessels: Spectral non-stationarity arises from switching between transit and active trawling modes. Winch and mechanical gear noise introduce additional tonal components absent in other classes.

Class D — Passengers/Ferries: Large vessels with fixed regular routes and consistent engine loads. Stable harmonic series dominates 50–500 Hz. May be confused with Class B due to overlapping blade-rate range; separation relies on source level and harmonic spacing differences.

Class E — Ocean Liners/Tugboats: Highest source levels in the dataset. Very low fundamental frequencies result from large, slow-turning propellers. Multi-shaft configurations produce complex overlapping harmonic patterns. Cavitation is present at full operational load.

VI. Formal Problem Definition

A. Input Space

Let $\mathbf{x} \in \mathbb{R}^N$ denote a fixed-length audio segment captured by the hydrophone, where:

$$N = T_s \times f_s, \tag{5}$$

T_s is the segment duration in seconds (e.g., 3 s) and f_s is the (resampled) sampling rate. Each segment is obtained by applying a sliding window with hop size $H < T_s$ to a raw recording, discarding any terminal remainder shorter than T_s.

B. Label Space and Classification Objective

Each segment is assigned a discrete class label y from the finite set:

$$y \in \mathcal{Y} = \{A, B, C, D, E\}. \tag{6}$$

The learning objective is to find an optimal classifier f^* that minimises the expected classification loss $\mathcal{L}$ over the joint data distribution $p(\mathbf{x}, y)$:

$$f^* = \arg\min_f \mathbb{E}_{(\mathbf{x}, y)\sim p}[\mathcal{L}(f(\mathbf{x}), y)]. \tag{7}$$

In practice, f operates on a compact feature representation $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^D$ with $D \ll N$, extracted via handcrafted signal processing (MFCCs, LOFAR, DEMON) or a learned convolutional encoder (log-mel spectrogram).

C. Key Assumptions and Constraints

The following assumptions govern the problem formulation:

1) **Single-channel input:** Each segment is captured by one omnidirectional hydrophone; no spatial beamforming is assumed.
2) **Mutual exclusivity:** Labels are exhaustive and mutually exclusive — each segment belongs to exactly one class in $\mathcal{Y}$.
3) **Recording-level splitting:** Train, validation, and test sets are partitioned at the *recording* level (not the segment level) to prevent data leakage between overlapping windows derived from the same source file.
4) **Class imbalance:** ShipsEar is imbalanced; model training uses class-weighted loss, and evaluation relies on macro-averaged F1-score and AUC-ROC rather than raw accuracy alone.
5) **Reproducibility:** All random operations (splitting, weight initialisation) are seeded with fixed values from $\{0, 7, 21, 42, 99\}$; mean $\pm$ standard deviation is reported across five runs.

D. Performance Metrics

Let $\hat{y} = f(\mathbf{x})$ denote the predicted label. The primary evaluation metrics are:

$$\text{Accuracy} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbf{1}[\hat{y}_i = y_i], \tag{8}$$

$$\text{Macro-F1} = \frac{1}{|\mathcal{Y}|} \sum_{c \in \mathcal{Y}} \frac{2 P_c R_c}{P_c + R_c}, \tag{9}$$

$$\text{Macro-AUC} = \frac{1}{|\mathcal{Y}|} \sum_{c \in \mathcal{Y}} \text{AUC}_{\text{OVR}}(c), \tag{10}$$

where P_c and R_c are the precision and recall for class c, $\mathcal{T}$ is the held-out test set, and AUC_{OVR} is computed under a one-vs-rest scheme.

VII. Conclusion

This part has established the complete scientific context required before any data processing or modelling. The ShipsEar dataset provides 90 hydrophone recordings across five acoustically distinct vessel classes in a realistic shallow-water environment. The dominant noise mechanisms — propeller cavitation, LOFAR engine harmonics, and blade-rate modulation — motivate the feature engineering choices pursued in Parts 3–4. The formal problem definition introduces the mathematical notation, loss framework, evaluation metrics, and experimental constraints used throughout the remainder of the project.

References

[1] D. Santos-Dom´ınguez, S. Torres-Guijarro, A. Cardenal-Lo´pez, and A. Pena, "ShipsEar: An underwater vessel noise database," *Applied Acoustics*, vol. 113, pp. 64–69, 2016. https://doi.org/10.1016/j.apacoust.2016.06.008

[2] R. J. Urick, *Principles of Underwater Sound*, 3rd ed. New York, NY, USA: McGraw-Hill, 1983.

[3] D. Ross, *Mechanics of Underwater Noise*. New York, NY, USA: Pergamon Press, 1976.

[4] K. V. Mackenzie, "Nine-term equation for sound speed in the oceans," *Journal of the Acoustical Society of America*, vol. 70, no. 3, pp. 807–812, 1981.